```
"C:\Users\nicop\OneDrive\Documentos\Universidad\Proyecto de Grado STREDEMIA\STREDEMIA\.venv\Scripts\python.exe" "C:\Users\nicop\OneDrive\Documentos\Universidad\Proyecto de Grado STREDEMIA\STREDEMIA\.venv\Modelos\DNN_Model.py"
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1776287793.142725   14548 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
WARNING: All log messages before absl::InitializeLog() is called are written to STDERR
I0000 00:00:1776287796.029537   14548 port.cc:153] oneDNN custom operations are on. You may see slightly different numerical results due to floating-point round-off errors from different computation orders. To turn them off, set the environment variable `TF_ENABLE_ONEDNN_OPTS=0`.
Entrenamiento: (381, 33)
Validación:    (37, 33)
Prueba:        (37, 33)
Variables de entrada (features): 33
  - Ítems SISCO:     21
  - Variables contextuales: 12
Scaler guardado en carpeta 'Modelos'

inicio de validación cruzada con 5 folds (pliegues)...

*-*-*-* Fold 1/5
I0000 00:00:1776287797.913001   14548 cpu_feature_guard.cc:227] This TensorFlow binary is optimized to use available CPU instructions in performance-critical operations.
To enable the following instructions: SSE3 SSE4.1 SSE4.2 AVX AVX2 AVX_VNNI FMA, in other operations, rebuild TensorFlow with the appropriate compiler flags.
WARNING:tensorflow:TensorFlow GPU support is not available on native Windows for TensorFlow >= 2.11. Even if CUDA/cuDNN are installed, GPU will not be used. Please use WSL2 or the TensorFlow-DirectML plugin.
F1-score en fold 1: 0.9652
*-*-*-* Fold 2/5
F1-score en fold 2: 0.9291
*-*-*-* Fold 3/5
WARNING:tensorflow:5 out of the last 7 calls to <function TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_distributed at 0x0000028C1B69B7E0> triggered tf.function retracing. Tracing is expensive and the excessive number of tracings could be due to (1) creating @tf.function repeatedly in a loop, (2) passing tensors with different shapes, (3) passing Python objects instead of tensors. For (1), please define your @tf.function outside of the loop. For (2), @tf.function has reduce_retracing=True option that can avoid unnecessary retracing. For (3), please refer to https://www.tensorflow.org/guide/function#controlling_retracing and https://www.tensorflow.org/api_docs/python/tf/function for  more details.
WARNING:tensorflow:6 out of the last 9 calls to <function TensorFlowTrainer.make_predict_function.<locals>.one_step_on_data_distributed at 0x0000028C1B69B7E0> triggered tf.function retracing. Tracing is expensive and the excessive number of tracings could be due to (1) creating @tf.function repeatedly in a loop, (2) passing tensors with different shapes, (3) passing Python objects instead of tensors. For (1), please define your @tf.function outside of the loop. For (2), @tf.function has reduce_retracing=True option that can avoid unnecessary retracing. For (3), please refer to https://www.tensorflow.org/guide/function#controlling_retracing and https://www.tensorflow.org/api_docs/python/tf/function for  more details.
F1-score en fold 3: 0.9528
*-*-*-* Fold 4/5
F1-score en fold 4: 0.9285
*-*-*-* Fold 5/5
F1-score en fold 5: 0.9643

*-*-*-* Resumen validación cruzada *-*-*-*-*
  Fold 1: F1-macro = 0.9652
  Fold 2: F1-macro = 0.9291
  Fold 3: F1-macro = 0.9528
  Fold 4: F1-macro = 0.9285
  Fold 5: F1-macro = 0.9643
  Promedio: 0.9480
  Desv. estándar: 0.0163

*-*-*-*- Entrenamiento de modelo final *-*-*-*-*-
Epoch 1/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 1s 22ms/step - accuracy: 0.5359 - loss: 0.9521 - val_accuracy: 0.6486 - val_loss: 0.8054
Epoch 2/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.7560 - loss: 0.6504 - val_accuracy: 0.6757 - val_loss: 0.6659
Epoch 3/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.8110 - loss: 0.4757 - val_accuracy: 0.7027 - val_loss: 0.5787
Epoch 4/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.8349 - loss: 0.3946 - val_accuracy: 0.7838 - val_loss: 0.5226
Epoch 5/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.8493 - loss: 0.3307 - val_accuracy: 0.7838 - val_loss: 0.4915
Epoch 6/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.8732 - loss: 0.3041 - val_accuracy: 0.8108 - val_loss: 0.4370
Epoch 7/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.8565 - loss: 0.2894 - val_accuracy: 0.7838 - val_loss: 0.4189
Epoch 8/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.8995 - loss: 0.2331 - val_accuracy: 0.7838 - val_loss: 0.4037
Epoch 9/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9234 - loss: 0.2156 - val_accuracy: 0.8108 - val_loss: 0.3472
Epoch 10/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9282 - loss: 0.1977 - val_accuracy: 0.8378 - val_loss: 0.3082
Epoch 11/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9139 - loss: 0.2307 - val_accuracy: 0.8378 - val_loss: 0.3092
Epoch 12/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9354 - loss: 0.1532 - val_accuracy: 0.8919 - val_loss: 0.2768
Epoch 13/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9402 - loss: 0.1596 - val_accuracy: 0.8649 - val_loss: 0.2884
Epoch 14/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9474 - loss: 0.1589 - val_accuracy: 0.8649 - val_loss: 0.2787
```

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Epoch 15/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.9354 - loss: 0.1742 - val_accuracy: 0.8919 - val_loss: 0.2553
Epoch 16/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9378 - loss: 0.1519 - val_accuracy: 0.8919 - val_loss: 0.2748
Epoch 17/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9545 - loss: 0.1179 - val_accuracy: 0.9459 - val_loss: 0.2154
Epoch 18/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9569 - loss: 0.1168 - val_accuracy: 0.9189 - val_loss: 0.1875
Epoch 19/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9641 - loss: 0.1261 - val_accuracy: 0.9189 - val_loss: 0.1927
Epoch 20/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9545 - loss: 0.1149 - val_accuracy: 0.9459 - val_loss: 0.1658
Epoch 21/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9593 - loss: 0.1358 - val_accuracy: 0.9459 - val_loss: 0.1722
Epoch 22/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9737 - loss: 0.0843 - val_accuracy: 0.9730 - val_loss: 0.1704
Epoch 23/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9617 - loss: 0.0988 - val_accuracy: 0.9459 - val_loss: 0.1557
Epoch 24/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9522 - loss: 0.1259 - val_accuracy: 0.9459 - val_loss: 0.1405
Epoch 25/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9785 - loss: 0.0714 - val_accuracy: 0.9459 - val_loss: 0.1333
Epoch 26/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9641 - loss: 0.0808 - val_accuracy: 0.9730 - val_loss: 0.1228
Epoch 27/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9737 - loss: 0.0807 - val_accuracy: 0.9459 - val_loss: 0.1385
Epoch 28/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9641 - loss: 0.0787 - val_accuracy: 0.9730 - val_loss: 0.1226
Epoch 29/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 12ms/step - accuracy: 0.9809 - loss: 0.0654 - val_accuracy: 0.9730 - val_loss: 0.1320
Epoch 30/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9833 - loss: 0.0693 - val_accuracy: 0.9730 - val_loss: 0.1535
Epoch 31/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9737 - loss: 0.0546 - val_accuracy: 0.9459 - val_loss: 0.1285
Epoch 32/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9785 - loss: 0.0613 - val_accuracy: 0.9730 - val_loss: 0.1187
Epoch 33/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9809 - loss: 0.0564 - val_accuracy: 0.9730 - val_loss: 0.1128
Epoch 34/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9713 - loss: 0.0674 - val_accuracy: 0.9730 - val_loss: 0.1227
Epoch 35/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9833 - loss: 0.0389 - val_accuracy: 0.9730 - val_loss: 0.1223
Epoch 36/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9833 - loss: 0.0413 - val_accuracy: 0.9730 - val_loss: 0.0984
Epoch 37/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9856 - loss: 0.0510 - val_accuracy: 0.9730 - val_loss: 0.0894
Epoch 38/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9904 - loss: 0.0364 - val_accuracy: 0.9730 - val_loss: 0.0783
Epoch 39/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9713 - loss: 0.0654 - val_accuracy: 0.9730 - val_loss: 0.1177
Epoch 40/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9809 - loss: 0.0522 - val_accuracy: 0.9730 - val_loss: 0.0954
Epoch 41/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9856 - loss: 0.0472 - val_accuracy: 0.9730 - val_loss: 0.0749
Epoch 42/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9904 - loss: 0.0349 - val_accuracy: 0.9730 - val_loss: 0.0690
Epoch 43/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9904 - loss: 0.0383 - val_accuracy: 0.9730 - val_loss: 0.0568
Epoch 44/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9809 - loss: 0.0706 - val_accuracy: 0.9730 - val_loss: 0.0860
Epoch 45/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9880 - loss: 0.0303 - val_accuracy: 0.9730 - val_loss: 0.0769
Epoch 46/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 11ms/step - accuracy: 0.9737 - loss: 0.0513 - val_accuracy: 0.9730 - val_loss: 0.0463
Epoch 47/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 10ms/step - accuracy: 0.9904 - loss: 0.0321 - val_accuracy: 0.9730 - val_loss: 0.0835
Epoch 48/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9880 - loss: 0.0334 - val_accuracy: 0.9730 - val_loss: 0.0628
Epoch 49/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9952 - loss: 0.0213 - val_accuracy: 0.9730 - val_loss: 0.0343
Epoch 50/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9904 - loss: 0.0209 - val_accuracy: 1.0000 - val_loss: 0.0170
Epoch 51/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9880 - loss: 0.0441 - val_accuracy: 1.0000 - val_loss: 0.0186
Epoch 52/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9856 - loss: 0.0290 - val_accuracy: 0.9730 - val_loss: 0.0374
Epoch 53/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9976 - loss: 0.0155 - val_accuracy: 0.9730 - val_loss: 0.0431
Epoch 54/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 9ms/step - accuracy: 0.9952 - loss: 0.0166 - val_accuracy: 0.9730 - val_loss: 0.0418
Epoch 55/150
14/14 ━━━━━━━━━━━━━━━━━━━━ 0s 8ms/step - accuracy: 0.9928 - loss: 0.0291 - val_accuracy: 0.9730 - val_loss: 0.0330

*-*-*-*- Evaluación final sobre datos de prueba *-*-*-*-

Reporte de clasificación:
              precision    recall  f1-score   support

        bajo       0.75      1.00      0.86         3
```

```
        medio       0.96      0.86      0.91        28
         alto       0.62      0.83      0.71         6

     accuracy                           0.86        37
    macro avg       0.78      0.90      0.83        37
 weighted avg       0.89      0.86      0.87        37

Modelo entrenado con éxito...

*-*-*-* MÉTRICAS Y EVALUACIONES *-*-*-*-*

AUC-ROC
Macro (one-vs-rest): 0.9668

Por clases:
  AUC-ROC bajo: 1.0000
  AUC-ROC medio: 0.9325
  AUC-ROC alto: 0.9677
Curvas ROC por clases archivadas en carpeta 'Resultados'
Matriz de confusión almacenada en carpeta 'Resultados'
Curvas de entrenamiento guardadas en carpeta 'Resultados'

Modelo ejecutado con éxito.
Métricas y evaluaciones generadas con éxito
Modelo Guardado (Carpeta Modelos)

Process finished with exit code 0
```